Seminar

Sisteme de ecuații liniare

$x \in K^n : Ax = b, \; A \in M_{m,n}(K), \; b \in K^m$

Teoremă: Sistemul (1) este compatibil $\Leftrightarrow$ rang A = rang $\bar{A}$

ex: a) $\begin{cases} 3x_1 + 4x_2 + x_3 + 2x_4 = 3 \\ 6x_1 + 8x_2 + 2x_3 + 5x_4 = 7 \\ 9x_1 + 12x_2 + 3x_3 + 10x_4 = 13 \end{cases}$, $(x_1, x_2, x_3, x_4) \in \mathbb{R}^4$

$$\bar{A} = \left(\begin{array}{cccc|c} 3 & 4 & 1 & 2 & 3 \\ 6 & 8 & 2 & 5 & 7 \\ 9 & 12 & 3 & 10 & 13 \end{array}\right)$$

$\begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = 1 \neq 0 \Rightarrow$ rang $A \geq 2$

$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 10 & 9 \end{vmatrix} = 0$; $\begin{vmatrix} 1 & 2 & 4 \\ 2 & 5 & 8 \\ 3 & 10 & 12 \end{vmatrix} = 0 \Rightarrow$ rang $A = 2$

$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 10 & 13 \end{vmatrix} = 0 \Rightarrow$ rang $\bar{A} = 2$

$\Rightarrow$ sistemul este compatibil nedeterminat (subdeterminat)

$\begin{cases} x_3 + 2x_4 = 3 - 3x_1 - 4x_2 \; | (-2) \\ 2x_3 + 5x_4 = 7 - 6x_1 - 8x_2 \end{cases}$

$\Rightarrow x_4 = 1$

$x_3 = 1 - 3x_1 - 4x_2$

Mulțimea soluțiilor

$$\{(x_1, x_2, 1-3x_1-4x_2, 1),\ x_1, x_2 \in \mathbb{R}\}$$

b) $\begin{cases} x_1 + x_2 - 3x_3 = -1 \\ 2x_1 + x_2 - 2x_3 = 1 \\ x_1 + x_2 + x_3 = 3 \\ x_1 + 2x_2 - 3x_3 = 1 \end{cases}$ în $\mathbb{R}^3$, $\bar{A} = \left(\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 2 & 1 & -2 & 1 \\ 1 & 1 & 1 & 3 \\ 1 & 2 & -3 & 1 \end{array}\right)$ $(x_1, x_2, x_3) \in \mathbb{R}^3$

$$\begin{vmatrix} 1 & 1 & -3 \\ 2 & 1 & -2 \\ 1 & 1 & 1 \end{vmatrix} = 1 + \cancel{2} - \cancel{2} + 3 + \cancel{2} - \cancel{2} = 4 \neq 0 \Rightarrow$$

$$\text{rang } A \geq 3 \Rightarrow \text{rang } A = 3$$

$$\begin{vmatrix} 1 & 1 & -3 & -1 \\ 2 & 1 & -2 & 1 \\ 1 & 1 & 1 & 3 \\ 1 & 2 & -3 & 1 \end{vmatrix} \overset{c_1-c_2}{=} \begin{vmatrix} 0 & 1 & -3 & -1 \\ 1 & 1 & -2 & 1 \\ 0 & 1 & 1 & 3 \\ -1 & 2 & -3 & 1 \end{vmatrix} \overset{l_2+l_4}{=}$$

$$= \begin{vmatrix} 0 & 1 & -3 & -1 \\ 0 & 3 & -5 & 2 \\ 0 & 1 & 1 & 3 \\ -1 & 2 & -3 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -3 & -1 \\ 3 & -5 & 2 \\ 1 & 1 & 3 \end{vmatrix} \begin{matrix} c_2 - c_1 \\ = \\ c_3 - 3c_1 \end{matrix}$$

$$= \begin{vmatrix} 1 & -4 & -4 \\ 3 & -8 & -7 \\ 1 & 0 & 0 \end{vmatrix} = \begin{vmatrix} 4 & 4 \\ 8 & 7 \end{vmatrix} = -4 \neq 0 \Rightarrow \text{rang } \bar{A} = 4 \neq \underset{\text{rang } A}{3}$$

$\Rightarrow$ (S) incompatibil

ex: Să se rezolve sistemele

a) $\begin{cases} 5x_1-3x_2+2x_3+4x_4=3 \\ 4x_1-2x_2+3x_3+7x_4=1 \\ 8x_1-6x_2-x_3-5x_4=9 \\ 7x_1-3x_2+7x_3+17x_4=\alpha \end{cases}$ în $\mathbb{R}^4$, $\alpha\in\mathbb{R}$

$$\overline{A} = 0\neq\left(\begin{array}{cccc|c} \boxed{\begin{matrix}5 & -3 \\ 4 & -2\end{matrix}} & \begin{matrix}2 \\ 3\end{matrix} & \begin{matrix}4 \\ 7\end{matrix} & \begin{matrix}3 \\ 1\end{matrix} \\ 8 & -6 & -1 & -5 & 9 \\ 7 & -3 & 7 & 17 & \alpha \end{array}\right)$$

$$\begin{vmatrix} 5 & -3 & 2 \\ 4 & -2 & 3 \\ 8 & -6 & -1 \end{vmatrix} \overset{C_1+C_2}{=} \begin{vmatrix} 2 & -3 & 2 \\ 2 & -2 & 3 \\ 2 & -6 & -1 \end{vmatrix} = 2\begin{vmatrix} 1 & -3 & 2 \\ 1 & -2 & 3 \\ 1 & -6 & -1 \end{vmatrix} =$$

$$= 2\begin{vmatrix} 1 & -3 & 2 \\ 0 & 1 & 1 \\ 0 & -3 & -3 \end{vmatrix} = 2\begin{vmatrix} 1 & 1 \\ -3 & -3 \end{vmatrix} = 2(-3+3)=0$$

$$\begin{vmatrix} 5 & -3 & 2 \\ 4 & -2 & 3 \\ 7 & -3 & 7 \end{vmatrix} \overset{C_1+C_2}{=} \begin{vmatrix} 2 & -3 & 2 \\ 2 & -2 & 3 \\ 4 & -3 & 7 \end{vmatrix} = 2\begin{vmatrix} 1 & -3 & 2 \\ 1 & -2 & 3 \\ 2 & -3 & 7 \end{vmatrix} \underset{L_3-2L_1}{\overset{L_2-L_1}{=}}$$

$$= 2\begin{vmatrix} 1 & -3 & 2 \\ 0 & 1 & 1 \\ 0 & 3 & 3 \end{vmatrix} = 0$$

$$\begin{vmatrix} 5 & -3 & 4 \\ 4 & -2 & 7 \\ 8 & -6 & -5 \end{vmatrix} \overset{C_1+C_2}{=} \begin{vmatrix} 2 & -3 & 4 \\ 2 & -2 & 7 \\ 2 & -6 & -5 \end{vmatrix} = 2\begin{vmatrix} 1 & -3 & 4 \\ 1 & -2 & 7 \\ 1 & -6 & -5 \end{vmatrix} =$$

$$= 2\begin{vmatrix} 1 & -3 & 4 \\ 0 & 1 & 3 \\ 0 & -3 & -9 \end{vmatrix} = 2\begin{vmatrix} 1 & 3 \\ -3 & -9 \end{vmatrix} = 2(-9+9)=0$$

$$\begin{vmatrix} 5 & -3 & 4 \\ 4 & -2 & 7 \\ 7 & -3 & 17 \end{vmatrix} \overset{c_1+c_2}{=} \begin{vmatrix} 2 & -3 & 4 \\ 2 & -2 & 7 \\ 4 & -3 & 17 \end{vmatrix} = 2 \begin{vmatrix} 1 & -3 & 4 \\ 1 & -2 & 7 \\ 2 & -3 & 17 \end{vmatrix} \begin{matrix} l_2 - l_1 \\ = \\ l_3 - 2l_1 \end{matrix}$$

$$= 2 \begin{vmatrix} \; \end{vmatrix} = 0$$

$\Rightarrow \text{rang}\, A = 2$

$$m_c = \begin{vmatrix} 5 & -3 & 3 \\ 4 & -2 & 1 \\ 8 & -6 & 9 \end{vmatrix} \overset{c_1+c_2}{=} \begin{vmatrix} 2 & -3 & 3 \\ 2 & -2 & 1 \\ 2 & -6 & 9 \end{vmatrix} = 2 \begin{vmatrix} 1 & -3 & 3 \\ 1 & -2 & 1 \\ 1 & -6 & 9 \end{vmatrix} =$$

$$= 2 \begin{vmatrix} 1 & -3 & 3 \\ 0 & 1 & -2 \\ 0 & -3 & 6 \end{vmatrix} = 2 \begin{vmatrix} 1 & -2 \\ -3 & 6 \end{vmatrix} = 2(6-6) = 0$$

$$\begin{vmatrix} 5 & -3 & 3 \\ 4 & -2 & 1 \\ 7 & -3 & \alpha \end{vmatrix} \overset{c_1+c_2}{=} \begin{vmatrix} 2 & -3 & 3 \\ 2 & -2 & 1 \\ 2 & -3 & \alpha \end{vmatrix} = 2 \begin{vmatrix} 1 & -3 & 3 \\ 1 & -2 & 1 \\ 1 & -3 & \alpha \end{vmatrix} =$$

$$= 2 \begin{vmatrix} 1 & -3 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & \alpha-3 \end{vmatrix} = 2(\alpha-3) \begin{vmatrix} 1 & -3 \\ 0 & 1 \end{vmatrix} =$$

$$= 2(\alpha - 3) \;??\; \boxed{= 2\alpha}$$

C_I Dacă $\alpha \in \mathbb{R}^*$ atunci (S) este incompatibil

$(\text{rang}\, A = 2 \neq 3 = \text{rang}\, \overline{A})$

C_{II} Dacă $\alpha \in \{0\}$ atunci (S) compatibil nedeterminat

$$\begin{cases} 5x_1-3x_2=5-2x_3-4x_4 & \Big| 2 \\ 4x_1-2x_2=7-5x_3-6x_4 & \Big| (-3) \end{cases}$$

$$\overline{} \; (+)$$

$$-2x_1=3+5x_3+13x_4$$

$$x_1=-\frac{1}{2}(3+5x_3+13x_4)$$

$$\left.\begin{aligned} -2x_2&=1-3x_3-7x_4+6+10x_3+26x_4 \\ &=7+7x_3+14x_4 \end{aligned}\right\} \Rightarrow$$

$$\Rightarrow x_2=-\frac{1}{2}(7+7x_3+14x_4)$$

Soluțiile

$$\Rightarrow \left\{\left(-\frac{1}{2}(3+5x_3+13x_4), -\frac{1}{2}(7+7x_3+14x_4), x_3, x_4\right)\right\}, x_3, x_4 \in \mathbb{R}$$

b) $$\begin{cases} 2x_1-x_2+3x_3+4x_4=5 \\ 4x_1-2x_2+5x_3+6x_4=7 \\ 6x_1-3x_2+7x_3+8x_4=9 \\ \alpha x_1-4x_2+9x_3+10x_4=11 \end{cases}$$ în $\mathbb{R}^4$, $\alpha \in \mathbb{R}$

$$\overline{A}=\left(\begin{array}{cccc|c} 2 & -1 & 3 & 4 & 5 \\ 4 & -2 & 5 & 6 & 7 \\ 6 & -3 & 7 & 8 & 9 \\ \alpha & -4 & 9 & 10 & 11 \end{array}\right) \Rightarrow \begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix} = -2 \neq 0 \Rightarrow$$

rang $A \geq 2$

$$\begin{vmatrix} 3 & 4 & -1 \\ 5 & 6 & -2 \\ 7 & 8 & -3 \end{vmatrix} \overset{c_2-c_1}{=} \begin{vmatrix} 3 & 1 & -1 \\ 5 & 1 & -2 \\ 7 & 1 & -3 \end{vmatrix} \overset{\substack{l_2-l_1 \\ l_3-l_1}}{=} \begin{vmatrix} 3 & 1 & -1 \\ 2 & 0 & 1 \\ 4 & 0 & 2 \end{vmatrix} =$$

$$= -\begin{vmatrix} 2 & 1 \\ 4 & 2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 3 & 4 & 2 \\ 5 & 6 & 4 \\ 7 & 8 & 6 \end{vmatrix} \overset{c_2-c_1}{=} \begin{vmatrix} 3 & 1 & 2 \\ 5 & 1 & 4 \\ 7 & 1 & 6 \end{vmatrix} = 0$$

$$\begin{vmatrix} 3 & 4 & -1 \\ 5 & 6 & -2 \\ 9 & 10 & -4 \end{vmatrix} = \begin{vmatrix} 3 & 1 & -1 \\ 5 & 1 & -2 \\ 9 & 1 & -4 \end{vmatrix} \underset{l_3-l_1}{\overset{l_2-l_1}{=}} \begin{vmatrix} 3 & 1 & -1 \\ 2 & 0 & -1 \\ 6 & 0 & -3 \end{vmatrix} = 0$$

$$\begin{vmatrix} 3 & 4 & 2 \\ 5 & 6 & 4 \\ 9 & 10 & \alpha \end{vmatrix} \overset{c_2-c_1}{=} \begin{vmatrix} 3 & 1 & 2 \\ 5 & 1 & 4 \\ 9 & 1 & \alpha \end{vmatrix} \underset{l_3-l_1}{\overset{l_2-l_1}{=}} \begin{vmatrix} 3 & 1 & 2 \\ 2 & 0 & 2 \\ 6 & 0 & \alpha-2 \end{vmatrix} =$$

$$= -2\begin{vmatrix} 1 & 1 \\ 6 & \alpha-6 \end{vmatrix} = 16-2\alpha = 2(8-\alpha)$$

(I) Dacă $\alpha = 8$: $\text{rang}\, A = 2$

$$\begin{vmatrix} 3 & 4 & 5 \\ 5 & 6 & 7 \\ 7 & 8 & 9 \end{vmatrix} = \dots = 0 \; ; \begin{vmatrix} 3 & 4 & 5 \\ 5 & 6 & 7 \\ 9 & 10 & 11 \end{vmatrix} = 0$$

$\Rightarrow \text{rang}\, \overline{A} = 2 = \text{rang}\, A \Rightarrow$

$\Rightarrow (S)$ compatibil nedeterminat

$$\begin{cases} 3x_3+4x_4=5-2x_1+x_2 & \Big| (-3) \\ 5x_3+6x_4=7-4x_1+2x_2 & \Big| \cdot 2 \end{cases}$$
$\oplus$

$$x_3=-1-2x_1+x_2$$

$$4x_4=5-2x_1+x_2+3+6x_1-3x_2=$$

$$=8+4x_1-2x_2 \Rightarrow x_4=2+x_1-\frac{1}{2}x_2$$

Soluțiile sunt:

$$\left\{\left(x_1, x_2, -1-2x_1+x_2, 2+x_1-\frac{1}{2}x_2\right), x_1, x_2 \in \mathbb{R}\right\}$$

(II) Dacă $\alpha \neq 8$, $\alpha \in \mathbb{R} \setminus \{8\} \Leftrightarrow$ rang $A \geq 3$

$$\begin{vmatrix} 2 & -1 & 3 & 4 \\ 4 & -2 & 5 & 6 \\ 6 & -3 & 7 & 8 \\ \alpha & -4 & 9 & 10 \end{vmatrix} \overset{C_4-C_3}{=} \begin{vmatrix} 2 & -1 & 3 & 1 \\ 4 & -2 & 5 & 1 \\ 6 & -3 & 7 & 1 \\ \alpha & -4 & 9 & 1 \end{vmatrix} \begin{array}{l} L_2-L_1 \\ = \\ L_3-L_2 \\ L_4-L_1 \end{array}$$

$$= \begin{vmatrix} 2 & -1 & 3 & 1 \\ 2 & -1 & 2 & 0 \\ 2 & -1 & 2 & 0 \\ \alpha-2 & -3 & 6 & 0 \end{vmatrix} = 0 \Rightarrow \text{rang } A = 3$$

$$\begin{vmatrix} 2 & 3 & 4 & 5 \\ 4 & 5 & 6 & 7 \\ \alpha & 9 & 10 & 11 \\ 6 & 7 & 8 & 9 \end{vmatrix} = 0 \Rightarrow \text{rang } \bar{A} = 3 = \text{rang} A$$

$\Rightarrow$ (S) Comp. nedeterminat

$$\begin{cases} 2x_1+3x_3+4x_4=5+x_2 \\ 4x_1+5x_3+6x_4=7+2x_2 \\ \alpha x_1+9x_3+10x_4=11+4x_2 \end{cases}$$

$$\Delta = \begin{vmatrix} 2 & 3 & 4 \\ 4 & 5 & 6 \\ \alpha & 9 & 10 \end{vmatrix} = 2(8-\alpha)$$

$$\Delta_1 = \begin{vmatrix} 5+x_2 & 3 & 4 \\ 7+2x_2 & 5 & 6 \\ 11+4x_2 & 9 & 10 \end{vmatrix} = \dots \Rightarrow x_1 = \frac{\Delta_1}{\Delta}$$

$$\Delta_2 = \begin{vmatrix} 2 & 5+x_2 & 4 \\ 4 & 7+2x_2 & 6 \\ \alpha & 11+4x_2 & 10 \end{vmatrix} = \dots \Rightarrow x_2 = \frac{\Delta_2}{\Delta}$$

Soluțiile:

$$\left\{ \left(\frac{\Delta_1}{\Delta}, x_2, \frac{\Delta_2}{\Delta}, \frac{\Delta_3}{\Delta} \right) \right\}, x_2 \in \mathbb{R}$$

c) temă

$$\begin{cases} \alpha x_1 + x_2 + x_3 = 1 \\ x_1 + \alpha x_2 + x_3 = 1 \\ x_1 + x_2 + \alpha x_3 = 1 \end{cases} \quad \text{în } \mathbb{R}^3, \alpha \in \mathbb{R}$$